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Smaky 6 Emulator — User Guide

This guide covers how to build, run, and use the **Smaky 6 emulator**. For documentation about the Smaky 6 computer itself (commands, OS, hardware), see docs/SMAKY6_USER_GUIDE_EN.md.

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1. Requirements

Dependency	Minimum version	Purpose
CMake	3.16	Build system
SDL2	2.0	Window, keyboard, display
C compiler	C11 (gcc / clang)	Compilation

Dependency	Minimum version	Purpose
git	any	FetchContent for Z80 core

Optional (for PDF generation only):

Dependency	Purpose
pandoc	Markdown → PDF conversion
lualatex	PDF engine used by pandoc

2. Building

```
git clone https://github.com/your-username/smaky6emu
cd smaky6emu
cmake -B build -DCMAKE_BUILD_TYPE=Debug
cmake --build build
```

For a release build:

```
cmake -B build -DCMAKE_BUILD_TYPE=Release
cmake --build build
```

The resulting binary is build/smaky6emu.

3. ROM Files

Place ROM files in roms/ (relative to the repository root). The build system copies them automatically into build/roms/.

File	Size	Required	Description
roms/samos_sys17.rom	172 KB	Yes	Phantom bootloader ROM (SYS17 TMS2716)
roms/chargen.rom	2 KB	Optional	Character generator PROM

Note: If chargen.rom is absent, the emulator uses a built-in synthetic character table. Text will be legible but may differ slightly from the original hardware.

If samos_sys17.rom is missing, the emulator prints a warning and the CPU executes undefined memory — nothing useful will happen.

4. Quick Start

Boot into the machine monitor (no floppy):

```
cd build
./smaky6emu
```

Boot from a floppy image:

```
./smaky6emu -disk ../floppies/sys.img
```

Boot with two drives:

```
./smaky6emu -disk ../floppies/sys.img -disk2 ../floppies/data.img
```

Auto-boot and land at the SAMOS > prompt:

```
./smaky6emu -disk ../floppies/sys.img -autoboot
```

5. Command-Line Reference

Floppy drives

Option	Description
-disk 	Mount a floppy image on drive DX0:
-disk2 	Mount a floppy image on drive DX1:

Boot control

Option	Description
-autoboot	Inject Enter ~3 s after start to select floppy boot
-break-to-monitor	Inject SHIFT+BREAK to enter monitor mode at startup
-autoboot2 <n>	Second key code sent after autoboot (decimal or 0xHH; default 0x20 = Space)
-autoboot3 <n>	Optional third key code (default: disabled)
-autoboot-timeout <s>	Wall-clock timeout in autoboot mode (0 = off)

String injection

Option	Description
-inject-str <s>	Inject a string once the SAMOS > prompt is detected. Use \n for Enter.
-inject-delay <f>	Frames to wait after > prompt before injecting (default: 2)
-inject-via-fifo	Route -inject-str through the hardware keyboard FIFO instead of the fast path

Example — run LIST automatically:

```
./smaky6emu -disk ../floppies/sys.img -autoboot -inject-str "LIST\n"
```

Display

Option	Description
-vmode <m>	Force video mode: alpha (text only), graphic (graphics only), super (text + graphics)
-gfxbits 	Bitmap bit order: lsb (default) or msb
-scale <n>	Integer window scale 1–8 (default 2 → 1024 × 496 pixels)

Timing and timeouts

Option	Description
-timeout <s>	Global wall-clock timeout in seconds (0 = off; default 30 s when -trace is active)

Sound

Option	Default	Description
-no-beeper	—	Silence the machine's 1-bit buzzer (beeper is on by default)
-drive-sound	—	Enable synthesized floppy drive sounds: motor whir, head-step clicks, and sector-hole ticks (off by default)

6. Keyboard Mapping

The Smaky 6 has a QWERTZ Swiss keyboard layout with special function keys. The emulator maps them to standard PC keys as follows.

Special keys

PC key	Smaky 6 function
F1	CHANGE function key
F2	SEARCH function key
F3	SHOW function key
F4	COPY function key
F5	CURSOR function key
F6	PROGRA function key
F7	KILL function key
F11 or Pause	BREAK — triggers NMI → drops into SYSMON monitor

PC key	Smaky 6 function
Shift+F11 or Shift+Pause	SHIFT+BREAK — hard reset (reboots from DX0:)
Escape	Smaky ESC / line cancel
Tab	Inserts DX1: in the CLI command line

Standard keys

All printable ASCII characters are passed through directly. The Smaky 6 keyboard uses **QWERTZ** layout (Swiss German) — if your PC keyboard is QWERTY or AZERTY, some punctuation keys may differ.

7. Display Options

The Smaky 6 has two independent display layers:

- **Alpha layer** — 64 × 20 character text display
- **Graphic layer** — monochrome bitmap (>30 000 pixels)

The `-vmode` flag controls which layers are rendered:

Value	What is shown
alpha	Text layer only
graphic	Graphics layer only
super	Both layers overlaid (normal operation)

The `-scale` flag sets the integer zoom level. Scale 2 (the default) gives a 1024 × 496 window, comfortable on most monitors.

8. Floppy Images

Supported format

The emulator reads **Micropolis raw sector images**: 77 tracks × 16 sectors × 256 bytes = 315 392 bytes per disk.

Place image files anywhere and pass the path to `-disk` / `-disk2`. The `floppies/` directory in the repository is the conventional location.

Extracting files from a floppy

Use the Python tool included in `tools/`:

```
python3 tools/smaky6_fuse.py ../floppies/sys.dsk --list
python3 tools/smaky6_fuse.py ../floppies/sys.dsk --extract-all --out floppies/extracted/
```

9. Sound

The emulator reproduces two categories of sound through SDL2 audio output.

Buzzer (beeper)

The Smaky 6 has a simple 1-bit buzzer driven by port 0x03. Software produces tones by toggling the port in a tight loop; each write flips the speaker state. The emulator models this accurately with a unipolar square wave at 44 100 Hz.

The buzzer is **enabled by default**. Pass `-no-beeper` to silence it.

Floppy drive sounds

The emulator can synthesize the acoustic character of the Micropolis 5.25" hard-sectored drive:

- **Motor whir** — bandpass-filtered noise (300–1 500 Hz) that fades in when the spindle starts and fades out ~0.8 s after the last seek activity.
- **Head-step click** — a sharp crack (600–3 000 Hz bandpass noise with exponential decay) produced on every track seek step.
- **Sector-hole tick** — a soft noise burst fired once per sector hole as the disk rotates (16 ticks per revolution at ~300 RPM).

Drive sounds are **disabled by default** (synthesized sounds are a work-in-progress; real samples will be added later). Enable with `-drive-sound`.

Example — boot with beeper and drive sounds:

```
./smaky6emu -disk sys.img -autoboot -drive-sound
```

Example — mute everything:

```
./smaky6emu -disk sys.img -autoboot -no-beeper
```

10. Automation and Scripting

The emulator can be driven non-interactively by combining `-autoboot`, `-inject-str`, and `-timeout`.

Run a command and capture screen output:

```
./smaky6emu \  
-disk ../floppies/sys.img \  
-autoboot \  
-inject-str "LIST\n" \  
-timeout 20 \  
-screndump 2>screen.txt
```

Run with tracing for debugging:

```
./smaky6emu \  
-disk ../floppies/sys.img \  
-autoboot \  
-tracing
```

```
-trace \  
-timeout 15 2>trace.log
```

Dump RAM at the end of a run:

```
./smaky6emu \  
-disk ../floppies/sys.img \  
-autoboot \  
-inject-str "BASIC\n" \  
-timeout 30 \  
-dump-ram basic_init.bin
```

11. Debugging and Tracing

These flags are intended for emulator development and reverse-engineering. They produce output on **stderr**.

Flag	What it traces
-trace	Z80 program counter at key boot milestones
-traceflow	Control flow in low RAM after OS handoff
-tracekbd	Every keyboard status port read and CLA write
-tracesnd	Every write to port 0x03 (buzzer)
-trace08	All IN/OUT traffic on port 0x08
-trace11	All reads from port 0x11
-trace19	All writes to port 0x19 (floppy control)
-tracecd	All reads from port 0xCD (Winchester interface)
-tracefdc	Focused floppy ID/checksum stream events
-scremdump	Changed screen rows printed to stderr each frame

Tip: Combine with shell redirection to capture traces without mixing them with emulator output:

```
./smaky6emu -disk sys.img -autoboot -tracekbd 2>kbd.log
```

12. RAM Dumps

Automatic dump on exit:

```
./smaky6emu -disk sys.img -autoboot -timeout 10 -dump-ram snapshot.bin
```

Interactive dump during a running session:

Press Ctrl+D in the terminal, or send SIGUSR1 to the process:

```
kill -SIGUSR1 $(pgrep smaky6emu)
```

This writes a file named `smaky6_ram_NNNN_pcXXXX.bin` in the current directory, where `NNNN` is a sequence number and `XXXX` is the Z80 PC value at the time of the dump.

The 64 KB dump can be inspected with any hex editor or disassembler:

```
xxd snapshot.bin | less
objdump -b binary -m z80 -D snapshot.bin | less
```

13. Troubleshooting

Black screen / no video after boot

Try forcing a video mode:

```
./smaky6emu -disk sys.img -autoboot -vmode alpha
```

Graphics appear inverted or garbled

Try toggling the bitmap bit order:

```
./smaky6emu -disk sys.img -autoboot -gfxbits msb
```

Emulator exits immediately with error 043

The floppy image may be unreadable or in the wrong format. Check the file size: a valid image is exactly 315 392 bytes.

```
wc -c myimage.dsk
```

Keyboard input not reaching the emulator

Make sure the SDL window has focus (click on it). The emulator only processes keyboard events when its window is focused.

Autoboot does not reach the > prompt

Some floppy images require more time to load. Increase the injection delay:

```
./smaky6emu -disk sys.img -autoboot -inject-delay 100
```

How to generate / update the PDF manuals

```
./tools/generate_pdfs.sh
```